

# Decoding Accuracy Does Not Certify Plannability: An Optimizer-Conditioned Audit of World-Model Costs\*

Anonymous authors  
Paper under double-blind review

## Abstract

World-model planners do not consume average representation or decoding error; they optimize the ordering induced by a terminal cost on their own candidate distribution. We introduce an optimizer-conditioned cost audit that separates dynamics error, candidate availability, immediate selection, and adaptive search. On two contact-rich MetaWorld tasks and two released visual world-model checkpoints, every candidate is executed from a simulator snapshot and its endpoint is re-encoded, making candidate dynamics exact. With the same CEM budget, a simulator-state reference cost solves push in 64/64 episodes and pick-place in 49/64, whereas latent  $L_2$  solves 0/64 in every cell. A supervised state-decoding cost reaches at most 5/64 despite a 2.1 cm median object-decoding error on held-out expert frames. The reason emerges only under search: stateprobe/reference rank correlation falls from 0.43–0.55 on initial populations to 0.11–0.16 after refitting. On identical candidates, stateprobe selection incurs structured regret beyond a residual-matched noisy-score null; changing only the refitting cost then causes later populations to lose better physical candidates. Exact-success candidates are also sparse in the realized final populations. Thus apparently accurate average decoding can coexist with a cost that becomes unreliable on optimizer-induced inputs. The causal selection and refitting diagnosis is established for the fixed stateprobe composition; latent  $L_2$  is supported by the end-to-end exact-dynamics comparison.

## 1 Introduction

Visual world models increasingly support planning by predicting future latent states and minimizing their distance to a goal (Zhou et al., 2025; Terver et al., 2026). This decomposition creates at least three distinct failure sites: the predictor may produce the wrong future, the search may not propose a useful action sequence, or the terminal cost may prefer a physically worse sequence. Aggregate prediction or decoding error does not distinguish these cases. A planner consumes an *ordering* over its own candidates, often after the optimizer has moved those candidates away from the data on which the representation or readout was validated.

We ask: *if every candidate future is exact, when does a representation-derived cost still fail to support contact-rich planning?* In MetaWorld (Yu et al., 2020), we snapshot the simulator, execute each CEM candidate, render its endpoint, and pass that observation through the frozen encoder. This exact-dynamics construction preserves the real representation and terminal scorer while removing the learned predictor. Simulator state provides a reference score for every candidate.

We call the resulting procedure an *optimizer-conditioned cost audit*. It scores identical populations with both the deployed proxy and a physical reference, compares real selection with a matched-noise permutation null, and intervenes on the cost used for adaptive CEM refitting. A separate availability test asks whether an exactly successful sequence is present in the realized final population. The audit separates:

\*An LLM-based assistant was used for manuscript restructuring and language editing. The authors verified all scientific claims, analyses, and conclusions.

1. whether failure persists without learned-dynamics error;
2. whether the finite planner budget can solve the task with an informative cost;
3. whether useful candidates are present under the realized search; and
4. whether the proxy selects and preserves the better candidates that are available.

We contribute both the reusable audit and a controlled empirical diagnosis. First, exact dynamics plus latent or probe-derived cost remains near failure, while the same search budget succeeds with a simulator-state cost. Second, object decoding that appears accurate on held-out expert observations does not translate into optimizer-conditioned cost agreement: rank correlation and elite recall collapse during refitting. Third, the stateprobe misranks identical candidates beyond an equally noisy residual-matched baseline, and a cost-only intervention shows that this misranking degrades later proposals. Finally, we measure sparse exact-success candidate availability rather than silently attributing every failure to ranking. These mechanism tests apply to stateprobe; latent  $L_2$  is evaluated by the exact-dynamics terminal-cost comparison.

## 2 Related Work

**Latent world-model planning.** DINO-WM plans in the frozen feature space of a self-supervised visual encoder (Zhou et al., 2025). A subsequent TMLR study systematically evaluates design choices for joint-embedding predictive world models and releases the MetaWorld checkpoints and planning stack used here (Terver et al., 2026). Our experiments audit two of those released checkpoints; they do not make a claim about all JEPA systems. More broadly, a decision-centric view of world-model evaluation argues that predictive quality should be connected to downstream decision quality (Yu et al., 2026). We instantiate that connection at the optimizer–cost interface.

**Optimization against imperfect models and costs.** Model-based control has long studied policies that exploit learned-dynamics error (Janner et al., 2019) and the mismatch between prediction objectives and control performance (Lambert et al., 2020). More recent work shows that gradient-based planning can produce out-of-distribution or adversarial action trajectories and smooths the induced planning landscape through adversarial training (Parthasarathy et al., 2025). We instead hold dynamics exact and study the terminal scorer. Selection over noisy estimates already creates an optimizer’s curse (Smith & Winkler, 2006); proxy and target can additionally decouple under optimization (Gao et al., 2023; Skalse et al., 2022; Karwowski et al., 2024). Our matched-residual null measures how much candidate misranking remains beyond the generic noisy-selection effect.

**Closest concurrent diagnoses.** Concurrent work in latent planning addresses complementary interfaces. RC-aux reshapes a representation using a reachability objective (Li et al., 2026b); TRM learns a trajectory-ordering terminal metric (Li et al., 2026a); and ACID augments the terminal objective with inverse-dynamics consistency along a proposed trajectory (Seo et al., 2026). IMWM replaces learned dynamics by environment rollout and identifies finite-population proposal coverage as an important failure mode (Gao et al., 2026). These studies establish related failure modes, but leave open how to separate cost selection from candidate availability under a controlled optimizer intervention. Our audit provides that decomposition: exact dynamics, dual scoring of identical contact-rich candidates, a residual-matched null, a cost-only intervention on adaptive refitting, and explicit measurement of exact-success availability in one harness.

## 3 Experimental Design

### 3.1 Models, tasks, and planning protocol

We evaluate the released `dino_wm_metaworld` and `jepa_wm_metaworld` checkpoints through their public `EncPredWM` interface (Terver et al., 2026). Both use their released preprocessing and action normalization. The contact tasks are `mw-push` and `mw-pick-place`; `mw-reach` is used only as a free-space control. The goal

image is the scripted expert trajectory’s final frame. Episode success is the simulator’s task flag at the final environment step, not an any-step latch. Its object-to-goal distance radii are 5 cm for push and 7 cm for pick-place.

Unless otherwise stated, the mechanistic experiments use receding-horizon CEM (Rubinstein, 1997) with 100 candidates, a top-10% elite set, six refitting iterations, and action horizon  $H = 6$ . We denote this budget by  $100 \times 6$ . The simulator-state reference arm is a positive control for the planner and budget; it is not a deployable vision-only method.

### 3.2 Oracle dynamics and three terminal costs

At each replan, the oracle-dynamics harness saves the full MuJoCo state. For every action sequence  $a$ , it restores the same snapshot, executes  $a$  in the simulator, renders the terminal observation, and encodes that observation with the frozen visual encoder. Thus candidate transitions are exact, while the representation and terminal scorer remain unchanged.

We compare three costs:

1. **Latent**  $L_2$ , the released Euclidean distance between the encoded endpoint and goal.
2. **Stateprobe**, an interpretable privileged diagnostic that decodes object position  $\hat{o}$  and end-effector position  $\hat{h}$  with supervised probes on frozen endpoint tokens using simulator labels, then applies the planning cost

$$\hat{c}(a) = \|\hat{o}_T(a) - \hat{o}_g\|_2 + 0.5\|\hat{h}_T(a) - \hat{h}_T(a)\|_2. \quad (1)$$

3. **Simulator-state reference**, which applies the same form to simulator object and hand state:

$$c^*(a) = \|o_T(a) - o_g\|_2 + 0.5\|h_T(a) - o_T(a)\|_2. \quad (2)$$

The reference is a shaped physical cost, not task return or a claim of a unique “true” objective.

### 3.3 Selection, null, refitting, and availability estimands

For a population  $\mathcal{A}$  scored by stateprobe and simulator reference, we define within-population selection regret

$$R_{\text{sel}} = c^* \left( \arg \min_{a \in \mathcal{A}} \hat{c}(a) \right) - \min_{a \in \mathcal{A}} c^*(a). \quad (3)$$

It measures the opportunity lost by proxy selection within the sampled population. Optimism is  $o(a) = c^*(a) - \hat{c}(a)$ . We report optimism enrichment of the proxy-selected elite set relative to its population. Let  $e_o(a) = \|\hat{o}_T(a) - o_T(a)\|_2$  and let  $E_{\hat{c}} \subset \mathcal{A}$  be the 10 candidates with smallest stateprobe cost. The object-only elite error excess and optimism enrichment are

$$\Delta_{\text{err}} = \frac{1}{|E_{\hat{c}}|} \sum_{a \in E_{\hat{c}}} e_o(a) - \frac{1}{|\mathcal{A}|} \sum_{a \in \mathcal{A}} e_o(a), \quad (4)$$

$$\Delta_{\text{opt}} = \frac{1}{|E_{\hat{c}}|} \sum_{a \in E_{\hat{c}}} o(a) - \frac{1}{|\mathcal{A}|} \sum_{a \in \mathcal{A}} o(a). \quad (5)$$

The first diagnostic concerns object decoding only; hand error enters the composite stateprobe cost and is reported separately in Appendix B.

The matched-residual null writes  $\hat{c}(a) = c^*(a) + r(a)$  and randomly permutes the observed residual multiset within each population before selecting an argmin. It preserves candidate states, reference costs, marginal error magnitudes, and search budget while breaking residual–candidate association. We use 1,000 permutations.

To test adaptive refitting, we fork CEM from candidate-for-candidate identical first populations. One branch refits on  $\hat{c}$  and the other on  $c^*$ . Subsequent samples use the same base Gaussian noise transformed by each

branch’s own fitted mean and variance. A true-state *carrier* supplies the shared sequence of receding-horizon simulator snapshots. The contrast is therefore a within-snapshot intervention on search.

Exact-success candidate availability is the fraction of replans whose final-iteration population contains at least one sequence satisfying the exact MetaWorld success predicate at its horizon endpoint. Selected success asks whether the proxy argmin is such a sequence. This is deliberately not called independent proposal coverage because the population is measured after earlier proxy-guided refits.

For the two contact tasks, physical task distance is the 3-D Euclidean object-to-goal distance  $d_{\text{task}}(a) = \|o_T(a) - o_g\|_2$ . Its within-population regret is

$$R_{\text{phys}} = d_{\text{task}}\left(\arg\min_{a \in \mathcal{A}} \hat{c}(a)\right) - \min_{a \in \mathcal{A}} d_{\text{task}}(a). \quad (6)$$

This differs from  $R_{\text{sel}}$ , which evaluates the composite shaped reference cost  $c^*$ .

### 3.4 Evaluation splits and uncertainty

The terminal-cost comparison uses fresh paired environment seeds 30000–30063. The same-population audit and matched null use independently fixed seeds 41000–41015; the branch intervention uses 42000–42007; and the availability audit uses 40000–40015. Confidence intervals for population-level estimands are percentile bootstraps clustered by environment seed (5,000 resamples). Episode rates use Wilson intervals, and paired episode contrasts use paired resampling or exact McNemar tests as appropriate. These reported evaluation ranges are disjoint from the eight off-policy probe-training environment seeds 20000–20007. Uncertainty is over environment seeds, not checkpoint-training seeds.

**Implementation integrity.** We corrected three environment-side mismatches before evaluation: renderer construction relative to camera assignment, the vertical image flip used in training preprocessing, and goal-frame selection. With the shipped latent  $L_2$  planner and its larger upstream budget, our Reach control succeeds in 6/16 episodes (37.5%, Wilson 95% interval [18.5, 61.4]%), compatible with the reported  $44.8 \pm 8.9\%$  but not a seed-matched replication (Terver et al., 2026). The exact audit protocol is given in Appendix A.

## 4 Results

### 4.1 Terminal-cost comparison under perfect dynamics

Table 1 varies only the terminal cost after replacing the learned predictor with simulator rollout. The same finite search solves nearly all push episodes and most pick-place episodes with the reference cost. Both representation-derived alternatives remain near failure.

Table 1: Held-out terminal-cost comparison on 64 fresh paired episodes per cell. Each task reports episode-end success and mean final object-to-goal distance in cm. Every row uses the same  $100 \times 6$  CEM budget and exact simulator candidate rollouts.

| dynamics  | repr.   | cost            | push         |             | pick-place   |             |
|-----------|---------|-----------------|--------------|-------------|--------------|-------------|
|           |         |                 | success      | dist.       | success      | dist.       |
| simulator | —       | state reference | <b>64/64</b> | <b>0.56</b> | <b>49/64</b> | <b>2.20</b> |
| simulator | DINO-WM | latent $L_2$    | 0/64         | 22.87       | 0/64         | 26.60       |
| simulator | DINO-WM | stateprobe      | 5/64         | 19.70       | 0/64         | 23.67       |
| simulator | JEPA-WM | latent $L_2$    | 0/64         | 22.87       | 0/64         | 26.68       |
| simulator | JEPA-WM | stateprobe      | 1/64         | 20.99       | 1/64         | 24.19       |

Wilson 95% intervals are [94.3, 100]% for reference push, [64.9, 85.3]% for reference pick-place, [3.4, 17.0]% for the 5/64 cell, [0.3, 8.3]% for each 1/64 cell, and [0, 5.7]% for each zero cell. The paired outcomes are

completely separated: all 64 push episodes and 49 pick-place episodes solved by the reference cost fail under latent  $L_2$ . The distance columns show that 0/64 is not a near-threshold floor: latent  $L_2$  ends roughly 23–27 cm from the goal, versus the 5 cm and 7 cm task radii.

Learned-dynamics error is therefore not required for failure in this harness, and the fixed search budget is capable of solving the tasks when supplied an informative cost. Separate sensitivity axes extending to 24 refits and to 300 candidates per iteration do not rescue the contact costs (Appendix C). Stateprobe performance reflects the complete representation–readout–cost–optimizer interface; the following experiments localize how that interface fails.

## 4.2 Held-out decoding does not predict ranking under search

Table 2 first evaluates the exact probes used by the planner on a held-out expert-trajectory split. Object localization appears strong in isolation: median Euclidean error is 2.1 cm, below both task distance radii. The hand readout is weaker, making clear that this is not a certificate for the full composite cost. The planning result is nevertheless striking: a readout with apparently useful object accuracy on ordinary held-out frames still supports at most 5/64 successful plans.

Table 2: Held-out expert-trajectory probe validation ( $n = 1,368$  transitions). RMSE is per coordinate and medians are Euclidean endpoint errors; all values are cm.

| representation | object RMSE | object median | hand RMSE | hand median |
|----------------|-------------|---------------|-----------|-------------|
| DINO-WM        | 1.71        | 2.09          | 4.31      | 6.65        |
| JEPA-WM        | 1.64        | 2.10          | 3.75      | 5.62        |

The relevant test is therefore the ranking on candidates produced by search. Figure 1 shows the shift directly. Stateprobe/reference Spearman agreement is 0.43–0.55 on initial populations but only 0.11–0.16 after five refits. Reference top-10% recall drops from 0.19–0.26 to 0.07–0.10, at or below the 0.10 expected overlap of two independent size-10 subsets of 100 candidates. Object error concurrently rises from 4.78–5.18 cm to 6.13–6.72 cm (Appendix B). Average held-out accuracy thus misses the optimizer-induced regime in which the cost is actually used.

## 4.3 The stateprobe cost misranks identical candidates

We next inspect the iteration-0 population at each audited replan, before adaptive refitting at that snapshot. Each of the same 100 action sequences is executed in MuJoCo and scored by both  $\hat{c}$  and  $c^*$ . Table 3 reports the selection diagnostics and matched null. First, candidates in the proxy top-10% have 1.24–1.40 cm more object decode error than the full population. Second, proxy elites have 2.16–2.33 cm more optimistic residuals than their populations, with all seed-clustered confidence intervals above zero. Third, proxy argmins incur 2.05–2.48 cm reference-cost regret within the sampled set—a substantial loss on tasks with 5 cm and 7 cm distance radii.

Table 3: Identical iteration-0 populations: 16 independent episodes, seven replans per episode, 112 populations, and 11,200 candidates per cell. Values are cm; brackets are seed-clustered 95% confidence intervals.  $\Delta_{\text{err}}$  is object-only elite error excess and  $\Delta_{\text{opt}}$  is optimism enrichment. The null permutes stateprobe residuals within each population. Null means omit intervals for compactness; actual-minus-null intervals include permutation and seed uncertainty.

| cell      | $\Delta_{\text{err}}$ | $\Delta_{\text{opt}}$ | actual $R_{\text{sel}}$ | null $R_{\text{sel}}$ | actual – null     |
|-----------|-----------------------|-----------------------|-------------------------|-----------------------|-------------------|
| DINO push | 1.27 [0.87, 1.64]     | 2.33 [2.08, 2.59]     | 2.48 [2.26, 2.69]       | 1.12                  | 1.36 [1.16, 1.55] |
| DINO pick | 1.25 [0.79, 1.76]     | 2.16 [1.81, 2.56]     | 2.25 [1.86, 2.64]       | 1.07                  | 1.17 [0.83, 1.54] |
| JEPA push | 1.24 [0.64, 1.83]     | 2.28 [1.88, 2.69]     | 2.05 [1.81, 2.32]       | 1.13                  | 0.92 [0.72, 1.13] |
| JEPA pick | 1.40 [0.88, 1.94]     | 2.21 [1.90, 2.59]     | 2.06 [1.76, 2.34]       | 1.15                  | 0.91 [0.66, 1.15] |

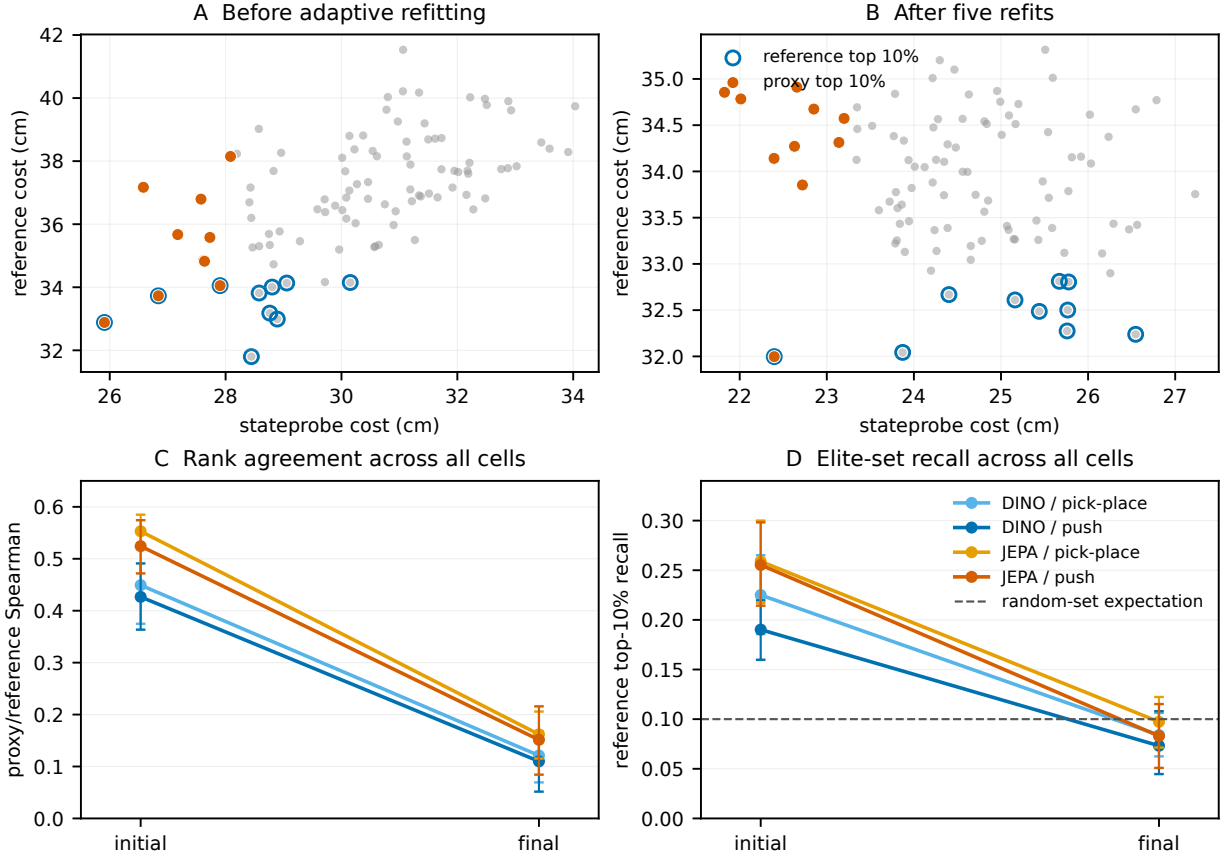


Figure 1: Optimizer-conditioned cost degradation. Panels A–B show all 100 candidates in the population fixed before plotting (DINO-WM push, seed 41000, first replan); colored points mark the proxy and reference top-10% sets. Panels C–D aggregate 112 populations and 11,200 candidates per stage in each of four checkpoint–task cells; error bars are seed-clustered 95% intervals over 16 episode seeds. Refitting reduces both rank agreement and recovery of the physically best elites.

Selection over any noisy score creates regret, and indeed the matched null incurs 1.07–1.15 cm. The actual score incurs roughly twice as much. Actual-minus-null regret is positive in all four cells, with lower confidence bounds between 0.66 and 1.16 cm. Marginal stateprobe error and the generic optimizer’s curse therefore explain part, but not all, of the observed misranking. The residuals retain candidate-specific structure that is harmful under selection.

#### 4.4 Changing only the refitting cost changes later proposals

Immediate misranking need not affect adaptive search. The shared-population branch intervention tests this link. Both branches begin with identical actions at the same snapshot; only the cost used to fit the next CEM distribution differs. Table 4 shows the probe-refit branch minus the reference-refit branch after five refits.

Table 4: Within-snapshot branch intervention ( $n = 8$  independent seeds). The final column is the difference in each branch’s best available simulator-state reference cost at iteration 5. Positive values favor reference-cost refitting. All intervals exclude zero; the seed-level mean difference is positive in 8/8 seeds in every cell (two-sided exact sign test  $p = 0.0078$  per cell).

| cell      | immediate $R_{\text{sel}}$ | final $\Delta$ best reference cost |
|-----------|----------------------------|------------------------------------|
| DINO push | 1.71 [1.22, 2.21] cm       | 1.64 [1.14, 2.20] cm               |
| DINO pick | 1.95 [1.33, 2.57] cm       | 1.51 [0.98, 2.06] cm               |
| JEPA push | 1.44 [0.97, 2.04] cm       | 2.01 [1.65, 2.35] cm               |
| JEPA pick | 1.65 [1.24, 2.05] cm       | 1.77 [0.96, 2.64] cm               |

After five refits, every probe-guided population has a worse *best* reference-cost candidate. Because both branches share the initial population and base sampling noise, the difference is caused by the intervened refitting cost within this search process. The direction is consistent in all 8/8 seed-level mean contrasts in every cell.

#### 4.5 Exact-success candidates are sparse under proxy-guided search

Table 5 evaluates the final CEM population across 112 replans from 16 episodes per cell. Exact successful candidates are rare: availability is 8.0% and 3.6% for DINO push and pick-place, and zero for both JEPA cells. At those replans, reranking the realized final population cannot select an exactly successful sequence when none is present. Selected physical-distance regret nevertheless remains positive on those same candidates in every cell.

Table 5: Final-iteration exact-success availability and selection over 112 replans from 16 independent episodes per cell. Brackets are seed-clustered 95% confidence intervals. Physical regret is  $R_{\text{phys}}$ , based on 3-D object-to-goal distance and distinct from the composite shaped-cost regret in Tables 3–4.

| cell      | exact-success availability | selected success | physical regret      |
|-----------|----------------------------|------------------|----------------------|
| DINO push | 8.0% [0.0, 17.0]           | 6.3% [0.0, 14.3] | 0.41 [0.25, 0.57] cm |
| DINO pick | 3.6% [0.0, 8.9]            | 1.8% [0.0, 5.4]  | 0.44 [0.24, 0.67] cm |
| JEPA push | 0.0%                       | 0.0%             | 0.42 [0.29, 0.55] cm |
| JEPA pick | 0.0%                       | 0.0%             | 0.59 [0.42, 0.75] cm |

The realized search therefore faces both sparse exact-success candidates and positive selection regret among the candidates it does produce. Because these populations follow earlier proxy refits, availability is a property of the whole realized search rather than the initial proposal distribution alone.

## 5 Discussion and Limitations

**Mechanistic picture.** The experiments expose a distributional failure that aggregate decoding metrics hide. The object readout is accurate on held-out expert frames, yet its full planning cost has only moderate rank agreement on fresh initial populations and nearly loses that ordering after refitting. Immediate misranking exceeds the matched-noise baseline, and the refitting intervention shows that it changes which physical candidates remain available. At the same time, exact-success candidates are sparse in final populations. The resulting diagnosis is a coupled selection-and-availability failure at the representation–readout–cost–optimizer interface.

**Implications.** A planning representation should be evaluated for the ordering it supports on the optimizer-induced distribution. When simulator state is available, we recommend the audit used here: dual-score identical populations, report rank agreement and within-population regret, compare against a residual-matched baseline, measure exact-success availability with its horizon and snapshot conditioning, and intervene on

adaptive refitting. This protocol distinguishes whether method development should prioritize cost learning, proposal generation, or both.

**Scope and limitations.** The audit covers MetaWorld, two contact tasks, two released checkpoints, one fixed stateprobe architecture and training seed, and CEM. The branch intervention uses eight seeds and a shared true-state snapshot carrier; uncertainty is across environment seeds rather than checkpoint-training seeds. The simulator-state cost and probe supervision are privileged diagnostics, not deployable methods. The mechanism tests establish misranking for the complete stateprobe composition—representation, readout, cost form, and optimizer—rather than for the encoder alone or latent  $L_2$ . The same-population regret is local to sampled candidates, and final availability also depends on previous refits, horizon, and encountered snapshots. Planner endpoint controls include MPPI and random shooting, but the causal mechanism has only been tested for CEM. Physical-robot, probe/weight sensitivity, and independent initial-proposal studies remain open.

## 6 Conclusion

Exact candidate dynamics do not rescue the two tested representation-derived costs, although the same finite search succeeds with an informative physical reference. The stateprobe explains why average decoding can be misleading: optimizer-induced candidates erode rank agreement, structured residuals alter immediate selection, and cost-only refitting changes later candidate quality. Sparse exact-success availability compounds this loss. World-model evaluation should therefore test a cost on the distribution and decisions induced by its optimizer; held-out prediction or decoding accuracy alone does not certify plannability.

### Broader Impact Statement

This work studies failure modes of simulation-based robot world-model planning and may support more reliable evaluation of learned planning systems. The experiments do not establish real-world robot safety or generality beyond the evaluated tasks, checkpoints, costs, and planners. Overgeneralizing the findings could create misplaced confidence in an audit or unwarranted conclusions about representation learning more broadly. Independent simulator and physical-system validation, together with deployment-specific safety controls, remains necessary before applying these conclusions to robots.

## References

- Baoqi Gao, Ruize Han, Miao Wang, and Song Wang. IMWM: Intuition models complement world models for latent planning. *arXiv preprint arXiv:2606.01626*, 2026.
- Leo Gao, John Schulman, and Jacob Hilton. Scaling laws for reward model overoptimization. In *International Conference on Machine Learning*, 2023.
- Michael Janner, Justin Fu, Marvin Zhang, and Sergey Levine. When to trust your model: Model-based policy optimization. In *Advances in Neural Information Processing Systems*, 2019.
- Jacek Karwowski, Oliver Hayman, Xingjian Bai, Klaus Kiendlhofer, Charlie Griffin, and Joar Skalse. Goodhart’s law in reinforcement learning. In *International Conference on Learning Representations*, 2024.
- Nathan Lambert, Brandon Amos, Omry Yadan, and Roberto Calandra. Objective mismatch in model-based reinforcement learning. In *Learning for Dynamics and Control*, 2020.
- Liangyu Li, Shengzhi Wang, and Qingwen Liu. Beyond Euclidean proximity: Repairing latent world models with horizon-matched trajectory reachability metrics. *arXiv preprint arXiv:2605.22164*, 2026a.
- Wenyuan Li, Guang Li, Keisuke Maeda, Takahiro Ogawa, and Miki Haseyama. Predictive but not plannable: RC-aux for latent world models. *arXiv preprint arXiv:2605.07278*, 2026b.



- Arjun Parthasarathy, Nimit Kalra, Rohun Agrawal, Yann LeCun, Oumayma Bounou, Pavel Izmailov, and Micah Goldblum. Closing the train-test gap in world models for gradient-based planning. *arXiv preprint arXiv:2512.09929*, 2025.
- Reuven Y. Rubinstein. Optimization of computer simulation models with rare events. *European Journal of Operational Research*, 99(1):89–112, 1997.
- Gawon Seo, Dongwon Kim, and Suha Kwak. ACID: Action consistency via inverse dynamics for planning with world models. *arXiv preprint arXiv:2607.02403*, 2026.
- Joar Skalse, Nikolaus Howe, Dmitrii Krashenninikov, and David Krueger. Defining and characterizing reward hacking. In *Advances in Neural Information Processing Systems*, 2022.
- James E. Smith and Robert L. Winkler. The optimizer’s curse: Skepticism and postdecision surprise in decision analysis. *Management Science*, 52(3):311–322, 2006. doi: 10.1287/mnsc.1050.0451.
- Basile Terver, Tsung-Yen Yang, Jean Ponce, Adrien Bardes, and Yann LeCun. What drives success in physical planning with joint-embedding predictive world models? *Transactions on Machine Learning Research*, 2026. URL <https://openreview.net/forum?id=cHZn5Gdh8e>.
- Tianhe Yu, Deirdre Quillen, Zhanpeng He, Ryan Julian, Karol Hausman, Chelsea Finn, and Sergey Levine. Meta-world: A benchmark and evaluation for multi-task and meta reinforcement learning. In *Conference on Robot Learning*, 2020.
- Yang Yu, Shiyuan Zhang, Yifei Sheng, Haoxiang Ren, and Haoxin Lin. How should world models be evaluated? a decision-making-centric position. *arXiv preprint arXiv:2606.15032*, 2026.
- Gaoyue Zhou, Hengkai Pan, Yann LeCun, and Lerrel Pinto. DINO-WM: World models on pre-trained visual features enable zero-shot planning. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pp. 79115–79135, 2025. URL <https://proceedings.mlr.press/v267/zhou25t.html>.

## A Audit Protocol and Reproducibility

**Candidate execution.** For every candidate action sequence, the harness restores an identical MuJoCo snapshot, executes the sequence, renders the endpoint, and runs the released encoder on that rendered observation. Simulator state is used only for evaluation, stateprobe supervision, or the explicitly privileged reference branch. Candidates are never assigned reference state by the learned predictor.

**Immediate selection.** The first-population audit evaluates 100 sequences per seed before any CEM refit. The top-10% elite sets are independently formed under stateprobe and reference cost. The residual null performs 1,000 within-population permutations. Seed-clustered percentile intervals use 5,000 bootstrap resamples.

**Adaptive branch.** Both CEM branches share the simulator snapshot, first candidate population, and base Gaussian noise. Each branch transforms later common noise by its own mean and variance. The reference-cost carrier determines the common sequence of MPC snapshots, so the audit isolates search behavior at those snapshots rather than comparing two closed-loop policies.

**Candidate availability.** Candidate-horizon success is evaluated with the simulator’s exact task predicate. It need not imply episode-end success after subsequent MPC replans. Intervals cluster replans by their environment seed. Because this is measured on final proxy-guided populations, it is not an independent estimate of the initial proposal distribution’s coverage.

Table 6: Fixed protocol used by the reported oracle-dynamics and mechanistic experiments.

| item                 | implementation                                                                                                                                                                                                                                      |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CEM distribution     | 4-D raw actions; initial mean 0 and variance 1; actions clipped to $[-1, 1]$ ; no warm start between replans                                                                                                                                        |
| Refitting            | 100 candidates, six iterations, 10 elites; full elite mean/variance update, no exponential smoothing; variance floor $10^{-4}$                                                                                                                      |
| Horizon and MPC      | $H = 6$ model steps; frame skip 5 gives 30 raw simulator actions per candidate; execute 3 model steps (15 raw actions) before replanning; horizon shrinks near episode end                                                                          |
| Episode protocol     | one zero-action environment warmup; at most 100 subsequent controlled raw actions; strict final-step simulator success flag                                                                                                                         |
| Environment and view | MetaWorld V3 goal-observable tasks; <b>corner2</b> camera at (0.75, 0.075, 0.7); $224 \times 224$ render; renderer reconstructed after camera assignment; vertical image flip                                                                       |
| Model inputs         | scripted expert’s final frame as goal; released visual preprocessing and released action normalization                                                                                                                                              |
| Object probe         | two-layer, six-head transformer with learned query, hidden size 512, 2.57M parameters; Adam, learning rate $5 \times 10^{-4}$ , batch 64, 8 epochs; final epoch used                                                                                |
| End-effector probe   | mean/max-pooled MLP, hidden size 512, 0.53M parameters; Adam, learning rate $10^{-3}$ , batch 64, 3 epochs; final epoch used                                                                                                                        |
| Probe data           | 12,312 expert transitions for training and 1,368 for validation, split by trajectory 90/10 with seed 0; 240 random-action frames from both contact tasks (seeds 20000–20007, 60 steps, sampled every 4 steps); off-policy-to-expert batch ratio 0.5 |
| Cost coefficient     | hand–object weight fixed at 0.5 in both stateprobe and reference costs before the confirmatory and mechanistic evaluation; not tuned on the reported evaluation seeds                                                                               |
| Uncertainty          | 5,000 seed-clustered bootstrap resamples; 1,000 within-population residual permutations; Wilson intervals and exact paired McNemar tests for episode rates                                                                                          |

## B Probe and Cost Details

The stateprobe decodes object and end-effector positions from frozen endpoint tokens. The object readout uses a two-layer transformer with a learned query token; the end-effector readout is a mean/max-pooled MLP. Both are trained with squared error on simulator-labelled off-policy frames, and no encoder parameters are updated. The stateprobe cost applies the same object-goal plus hand-object functional form as the simulator-state reference, comparing the decoded endpoint object with the decoded goal-frame object. Consequently, the audit diagnoses the full representation–readout–cost composition and cannot distinguish information absent from the encoder from information inaccessible to this readout. Held-out expert-trajectory errors for the exact probes are reported in Table 2.

Table 7: Stateprobe behavior on optimizer-induced candidates. Initial and final denote CEM iterations 0 and 5. Each row contains 112 populations and 11,200 candidates from 16 episode seeds; brackets are seed-clustered 95% intervals. Object and hand columns are mean Euclidean endpoint errors in cm.  $\rho$  compares stateprobe and reference shaped costs; reference top-10 recall is intersection-over-10 of their elite sets.

| repr. | task | stage   | object error      | hand error        | cost $\rho$       | reference top-10 recall |
|-------|------|---------|-------------------|-------------------|-------------------|-------------------------|
| DINO  | push | initial | 5.18 [4.76, 5.61] | 6.39 [5.76, 7.09] | 0.43 [0.36, 0.49] | 0.19 [0.16, 0.22]       |
| DINO  | push | final   | 6.72 [5.99, 7.43] | 6.60 [5.76, 7.50] | 0.11 [0.05, 0.16] | 0.07 [0.04, 0.11]       |
| DINO  | pick | initial | 4.98 [4.37, 5.57] | 6.16 [5.41, 7.03] | 0.45 [0.37, 0.52] | 0.22 [0.19, 0.27]       |
| DINO  | pick | final   | 6.35 [5.32, 7.35] | 6.27 [5.28, 7.43] | 0.12 [0.07, 0.17] | 0.08 [0.06, 0.11]       |
| JEPA  | push | initial | 4.79 [4.08, 5.51] | 8.52 [8.13, 9.03] | 0.52 [0.47, 0.57] | 0.26 [0.21, 0.30]       |
| JEPA  | push | final   | 6.13 [4.93, 7.34] | 8.70 [8.20, 9.33] | 0.15 [0.08, 0.22] | 0.08 [0.05, 0.12]       |
| JEPA  | pick | initial | 4.78 [4.15, 5.42] | 8.26 [7.92, 8.60] | 0.55 [0.52, 0.58] | 0.26 [0.22, 0.30]       |
| JEPA  | pick | final   | 6.15 [5.11, 7.15] | 8.38 [8.02, 8.74] | 0.16 [0.12, 0.21] | 0.10 [0.07, 0.12]       |

The expert split shows that the object readout is substantially more accurate than the hand readout. On the CEM populations actually used by the mechanism test, initial cost rank agreement is moderate ( $\rho = 0.43$ – $0.55$ ), but final-population agreement is only  $0.11$ – $0.16$  and reference top-10 recall is  $0.07$ – $0.10$ . These measurements characterize the fixed stateprobe composition; they do not establish invariance to a different readout architecture or training seed.

The probe weights use random-rollout episodes from environment seeds 20000–20007. Because those random-action frames are training augmentation, we do not present them as independent validation; Tables 2 and 7

instead use the held-out trajectory split and disjoint CEM evaluation seeds. The held-out terminal-cost comparison (30000–30063), selection audit (41000–41015), branch intervention (42000–42007), and availability audit (40000–40015) use disjoint environment seeds.

## C Search-Budget Sensitivity

A separate DINO-WM sensitivity run varies CEM refitting iterations on 16 fixed episodes (seeds 10000–10015) while keeping 100 candidates per iteration. Table 8 shows that increasing refits fourfold beyond the main setting does not rescue either contact cost, whereas the latent- $L_2$  Reach control saturates at 16/16 by six refits. At 12 refits, a separate population-size sweep over 50, 100, and 300 candidates gives push success 0/8, 0/8, and 0/8 for latent  $L_2$ , and 0/8, 1/8, and 0/8 for stateprobe. These supporting runs rule out a narrow tuning artifact over the tested range, not arbitrary search budgets.

Table 8: Episode-end success as CEM refitting iterations increase; each iteration contains 100 candidates and each cell has 16 paired episodes.

| task       | cost         | 2 refits | 6 refits | 12 refits | 24 refits |
|------------|--------------|----------|----------|-----------|-----------|
| push       | latent $L_2$ | 0/16     | 0/16     | 0/16      | 1/16      |
| push       | stateprobe   | 0/16     | 1/16     | 1/16      | 2/16      |
| pick-place | latent $L_2$ | 0/16     | 0/16     | 0/16      | 0/16      |
| pick-place | stateprobe   | 0/16     | 0/16     | 0/16      | 1/16      |
| reach      | latent $L_2$ | 5/16     | 16/16    | 16/16     | 16/16     |

## D Planner-Level Robustness Control

On DINO-WM push under oracle dynamics, stateprobe remains near failure with CEM (1/16), MPPI (0/16), and random shooting (2/16). Their free-space Reach controls are 16/16, 13/16, and 6/16, respectively. These endpoint controls show that the contact failure is not a CEM-only observation. They do not establish mechanism generality because the identical-population, matched-null, and refitting interventions were run only for CEM, and the planners have different effective search strength.